

EMILIA PROTOCOL · SEED

The accountability layer for the **agent economy**.

AI is about to take trillions of irreversible actions — moving money, deploying code, dispatching power, moving machines. **EMILIA is the portable, offline-verifiable proof that a named human authorized each one.** Antivirus scanned files. Firewalls filtered packets. EMILIA verifies actions.

THE INEVITABILITY

Software stopped asking permission. It just **acts** now.

Agents already release payments, push deploys, change permissions, curtail datacenters, and command robots — at machine speed, with no human in the loop. The blast radius is real and irreversible. Every regulator, insurer, and board is now asking the same question, and no one can answer it: **who authorized that, and can you prove it?**

THE GAP

**"A human approved this"
is a claim **no one can check.****

Today the proof lives in the operator's own **editable logs** — forgeable, backfillable, un-auditable by a regulator, court, or insurer without trusting the very party under review. Logs tell you what happened. They cannot prove who authorized it. **That evidence gap is a category.**

THE ARTIFACT

A receipt that proves a named human authorized **that exact action**.

TRUST-RECEIPT · VERIFIES OFFLINE

action release_payment \$40,000
approver jchen · Face ID ✓
bound sha256:9f2c... (exact action)
sig ed25519:R9w1...

✓ verified – no account, no server, no trust in us

No receipt, no execution. Change one byte → verification fails.

THE PRODUCT · THE CATEGORY

EMILIA Gate — the **Consequence Firewall**.

Deny-by-default at the actuator boundary. A consequential action runs only with a valid, in-scope, sufficiently-assured, non-replayed receipt — then emits proof it ran. A new security category: not antivirus (is this file bad?), not a firewall (is this packet allowed?) — **is this machine allowed to take this consequence, and can you prove who authorized it?**

request → **428** → **human signs exact action** → **verify** → **execute** → **execution receipt**

THE CATEGORY, VALIDATED

The market just funded the category. We're the **layer underneath it.**

In 2026 capital confirmed agent-action security is a category: **Arcade \$60M** (SYN Ventures · Morgan Stanley), **Oasis \$120M** (Craft · Sequoia · Accel), **GitGuardian \$50M** (Insight), Astrix acquired by Cisco. They secure *which machine* acts and *which tools* it may call. None produces the artifact regulators actually demand: **proof that a named human authorized this exact action.**

IDENTITY / NHI · OASIS, ASTRIX

which machine is acting

ACTION AUTH · ARCADE

the agent may call a tool

EMILIA

which **named human** authorized **this exact action** – verifiable offline

THE MARKET

Every consequential machine action is **addressable**.

TAM

\$100B+

AI trust, risk & security + agent governance by 2030–2035. Every irreversible action across 17+ verticals where a human must remain accountable. (analyst estimates)

SAM

\$20–40B

Regulated deployers where a missing approval is a **statutory** problem: finance, energy, healthcare, gov/defense, critical infra, robotics. (EP-addressable subset)

SOM · 3-YR BEACHHEAD

\$1–3B

The dev/agent-guard wedge + the managed control plane, landing first in energy (GRACE), finance, and DevOps change-control.

THE EXPANSION SURFACE

One protocol. **Seventeen verticals.** Same pattern.

Energy / Grid

Verifiable demand response — GRACE. Partner + federal path.

Financial services

Wires, beneficiary changes, AI trading — BEC & SOX.

Defense / autonomy

Human judgment over force — DoD 3000.09.

Healthcare

AI treatment / device overrides — FDA, EU Art. 14.

Autonomous vehicles

Emergency-maneuver authorization — NHTSA.

Production / DevOps

Deploy, migration, secret rotation — SOC 2 / SOX.

Robotics / industrial

Actuator-boundary signoff — ISO 10218.

Nuclear / critical infra

Two-person rule — NRC / NERC.

Space / satellite

Collision-avoidance maneuvers.

Maritime (MASS)

COLREGS decisions — IMO MASS Code.

Blockchain / DeFi

High-value smart-contract execution.

+ insurance, content, judicial, agriculture, aviation...

Same five conditions, same receipt.

Irreversible action · human accountable · third-party audit · proof outlives the operator. **That's the pattern. It's everywhere.**

WHY NOW

Regulation is converging on the artifact **we already ship.**

EU AI Act Art. 14/12 (human oversight + tamper-evident logs), NIST AI RMF, ISO 42001, eIDAS 2.0 qualified ledgers, FERC 2222 / NERC (energy), DoD 3000.09 (defense) — all demand verifiable human authorization and specify **no format for it**. EP is that format. There is a ~24-month window before a reference standard locks in. Whoever is the running, open, formally-verified standard when the rules land, wins.

THE MOAT

By the time a competitor starts,
we're **already the standard.**

Shipped, not slideware — the Gate, framework guards, a robot sidecar, three cross-language verifiers — running today.

Standards land-grab in flight — an IETF Internet-Draft cluster + a regulatory crosswalk. First-mover claims the namespace.

Offline-verifiable — works air-gapped and at the edge. Cloud-approval competitors structurally cannot follow.

Formally verified — 26 TLA+ invariants, 35 Alloy facts, 85 red-team cases. The assurance bar regulators require.

Network effect — once rails require receipts, every legitimate agent adopts issuance. That's how TLS won — and it doesn't reverse.

PROOF IT'S REAL

A standard, **with running code.**

4,220

automated tests passing

26

TLA+ safety invariants (TLC)

35

Alloy facts · 22 assertions

85

adversarial red-team cases

3

independent verifiers — JS · Python · Go

RR-1

conformance — earned, not asserted

IETF

Internet-Draft cluster (receipts · quorum · enforcement · ...)

Apache-2.0

open protocol — valid even if we vanish

BUSINESS MODEL

Open protocol for ubiquity. Operated trust root for revenue.

Free & open

The standard, the verifiers, the receipt profiles. Distribution and credibility; no lock-in fear.

Managed control plane

Hosted issuer, approver directory, transparency log, compliance/settlement evidence, audit dashboards.
~\$25K–\$150K ACV.

Who pays

Regulated deployers — never the government. The flywheel: regulator mandates the receipt → deployers must produce it → buy the managed layer.

Comps: Okta / Auth0 (identity) + Vanta / Drata (compliance) — CrowdStrike for machine actions.

GO TO MARKET

Wedge → **product** → **vertical** → **mandate.**

1 · Dev wedge

Receipt-Required for MCP / agent tools.
Developer-native enforcement; the crank
that seeds the Gate.

2 · The Gate

EMILIA Gate across clouds, APIs, robots.
The horizontal product + managed layer.

3 · Lighthouse vertical

GRACE (energy) with a committed partner
+ federal path. The first regulated, paid
reference.

4 · The standard

IETF + regulatory references →
procurement and insurance require EP. The
compounding endgame.

TRACTION & VALIDATION

Shipped. Partnered. In the standards room.

Product — EMILIA Gate + robot sidecar shipped & tested; three cross-language verifiers; RR-1 conformance.

Energy vertical — co-development partner (J Diesel NY) committed, MOU in motion, NETL/DOE unsolicited-proposal path.

Government — Utah Office of AI Policy engagement (third-party-auditor track).

Community — Lesson-18 human-authorization contribution to a Microsoft open course (community contribution, not endorsement).

Standards — IETF Internet-Draft cluster staged; coalition carrying EP at IETF agent2agent.

Honest stage: pre-revenue, pilot-next, building in the open — with a structural head start.

THE ASK

Raising a **\$2–3M seed** to turn a head start into the standard.

Standards-track

File & drive the I-D cluster to working-group adoption; lock the regulatory references.

First production pilot

GRACE observe-mode → enforce-mode with a real facility. The lighthouse case study.

The managed layer

Ship the control plane; first regulated-deployer revenue; 2–3 key hires.

Round size is milestone- and runway-based (~18–24 months). The re-rate: a shipped product, a committed vertical partner with a federal path, a standards land-grab in flight, and a defensible moat — not a pre-revenue protocol.

WHY I BUILT IT

Keep a human responsible for the future.

I built EMILIA to make trust between humans and AI something you can *prove* — to keep a named person accountable for the decisions that matter as machines start to act. I named it after my daughter, **Emilia** — because that's the future this protects. The window to own this layer is open now, and it won't be for long.

No receipt. No execution.

APPENDIX · SOURCES & METHODOLOGY

How the market is sized.

TAM — top-down

The AI trust, risk & security management + agent-governance category (Gartner coined "AI TRiSM"). Published analyst projections vary widely — roughly low-single-digit \$B in 2025 scaling toward \$100B+ by 2030–2035. Treat as directional, not precise.

SAM — bottom-up

EP's own estimate of regulated-deployer segments where a missing approval is a statutory problem: finance, energy, healthcare, gov/defense, critical infrastructure, robotics.

SOM — 3-yr obtainable

The dev/agent-guard wedge + managed control plane, landing first in energy (GRACE), finance, and DevOps change-control.

Demand drivers (regulatory): EU AI Act (Reg. 2024/1689) Arts. 12/14/17 · NIST AI RMF · ISO/IEC 42001 · eIDAS 2.0 (Reg. 2024/1183) · FERC Order 2222 · DoD Directive 3000.09.

Market & energy sources: IEA, "Energy and AI" (2025) — datacenter electricity ~415 TWh (2024) → ~945 TWh (2030) · Duke Nicholas Institute, "Rethinking Load Growth" (2025) — 76–126 GW flexible-load headroom at <1% curtailment · PJM (pays demand response) · Gartner (AI TRiSM category) · analyst projections (multiple firms; ranges vary).

EP metrics: tests, verifiers, TLA+/Alloy, red-team, and conformance are measured directly from the open-source codebase, not estimated.